Cybersecurity Risk Assessment & Policy Draft
For a University Organization

Cybersecurity Risk Assessment & Policy Draft

A Risk Assessment for a University Organization

Author: *Tanmay Madan*

South Asian University
New Delhi, India
Email: tanmay.madan@students.sau.ac.in

1. Introduction

In today's digital era, universities rely heavily on information technology systems to manage academic, administrative, and communication processes. From student information systems and online examination platforms to learning management systems and campus networks, these digital infrastructures store and process large volumes of sensitive data. This includes personal details of students and staff, academic records, financial information, and research data.

However, this increasing dependence on digital systems also makes universities attractive targets for cyberattacks. Unlike corporate environments, university networks are often open and decentralized, allowing access to a large number of users such as students, faculty, and external collaborators. This openness, combined with varying levels of cybersecurity awareness, creates multiple vulnerabilities that attackers can exploit.

This project focuses on conducting a cybersecurity risk assessment of a university IT system and proposing a comprehensive security policy. It also outlines an incident response plan and user awareness strategies to minimize risks and enhance the overall security posture of the institution.

2. Case Evidence

- **University of California San Francisco (2020)**
  The University of California San Francisco suffered a ransomware attack that encrypted critical research data. The attackers demanded a ransom, and the university paid approximately $1.14 million to regain access. This incident highlighted the vulnerability of academic research environments to cyber threats.
- **University of Greenwich (2016)**
  A data breach at the University of Greenwich exposed personal information of nearly 20,000 students due to a vulnerable website. The breach occurred because of poor security practices, leading to a fine by the UK Information Commissioner's Office.
- **Michigan State University (2020)**
  Michigan State University experienced a data breach that exposed sensitive information, including Social Security numbers and personal details of students and staff. The attack shows the risks associated with inadequate data protection and monitoring systems.

3. Objectives

- To identify major cybersecurity risks faced by universities such as phishing, malware, weak passwords, and unauthorized access.
- To study the impact of cyber threats on university systems, student records, and online services.
- To propose effective security policies related to password management, access control, network security, and data protection.
- To understand the importance of incident response planning and risk monitoring in managing cybersecurity threats.
- To promote cybersecurity awareness among students, faculty, and staff through safe cybersecurity practices and awareness measures.

4. Organization Chosen

University

A university environment has been chosen for this project because universities manage large amounts of sensitive data such as student records, examination details, faculty information, financial data, and research documents. Universities also have large networks with thousands of users including students, teachers, and staff, making them common targets for cyberattacks. Due to open internet access and shared systems, universities face several cybersecurity challenges that require proper security policies and risk management. Furthermore, chose this

topic because, as university students ourselves, we are directly connected to the digital systems used in educational institutions every day. From accessing learning management systems and online examination portals to using campus Wi-Fi and
cloud-based platforms, we regularly interact with technologies that can be vulnerable to cyber threats.

5. Risk Assessment Matrix

Risk	Description	Impact	Likelihood
Unauthorized Access to Student Portals	Attackers may gain access to student or faculty accounts using stolen credentials or weak passwords.	High	Medium
Data Breach of Student and Faculty Records	Sensitive information such as marks, personal details, and financial records may be leaked or stolen.	High	Medium
Phishing Attacks on Students and Staff	Fake emails or websites may trick users into revealing passwords or confidential information.	High	High
Malware and Ransomware Attacks	Malicious software can infect university systems, damage files, or lock important data	High	Medium
Unsecured Campus Wi-Fi Networks	Open or weakly secured university Wi-Fi networks can allow attackers to intercept data or access systems.	Medium	High
Server Downtime and DDoS Attacks	University websites and online portals may become unavailable due to excessive malicious traffic.	High	Medium
Examination System Attacks	Attackers may attempt to manipulate online examination systems, grades, or attendance records.	High	Low

Insider Threats	Students or staff with authorized access may misuse university systems or leak sensitive information.	Medium	Medium

6. Security Policies

Security policies are important for protecting university systems, networks, and sensitive information from cyber threats. These policies help improve cybersecurity, reduce risks, and ensure safe use of university resources.

6.1 Password Security Policy

Strong passwords should be used to prevent unauthorized access to university accounts and systems.

Policy Guidelines

- Passwords should contain a combination of letters, numbers, and special characters.
- Passwords should be updated regularly.
- Users should avoid sharing passwords with others.
- Multi-Factor Authentication (MFA) should be enabled for important accounts.

Purpose

This policy helps protect systems from unauthorized access and brute-force attacks.

6.2 Access Control Policy

Access control policies ensure that only authorized individuals can access university systems, applications, and data.

Policy Guidelines

- Access rights should be assigned based on user roles and responsibilities.

- Students, faculty members, and administrators should have different levels of system access.
- Administrative privileges should only be provided to authorized personnel.
- User accounts that are inactive or no longer required should be disabled immediately.
- Systems should automatically log out inactive users after a certain period of inactivity.

Purpose

This policy minimizes unauthorized access and helps protect confidential student, faculty, and administrative data.

6.3 Network Security Policy

The university network must be protected against cyber threats such as malware, unauthorized access, and network attacks.

Policy Guidelines

- Firewalls should be installed and configured to monitor incoming and outgoing network traffic.
- Antivirus and anti-malware software must be regularly updated on all systems.
- University Wi-Fi networks should use secure encryption methods such as WPA2 or WPA3.
- Unauthorized devices should not be allowed to connect to the university network.

Purpose

This policy helps secure university networks and reduces the risk of malware infections, hacking attempts, and data interception.

6.4 Data Protection and Backup Policy

Universities store large amounts of sensitive information that must be protected from loss, theft, or unauthorized access.

Policy Guidelines

- Sensitive student and faculty data should be stored securely and protected using encryption where possible.
- Important data should be backed up regularly to secure storage systems.
- Backup copies should be tested periodically to ensure successful recovery.
- Access to confidential data should be limited to authorized users only.
- Data sharing should follow proper authorization procedures.

Purpose

This policy helps prevent data loss and ensures that important information can be recovered during cyber incidents or system failures.

6.5 User Awareness and Training Policy

Students, faculty, and staff should be educated about cybersecurity threats and safe online practices.

Policy Guidelines

- Conduct regular cybersecurity awareness sessions.
- Educate users about phishing attacks and password security.
- Encourage safe browsing and reporting of suspicious activities.

Purpose

This policy helps reduce cybersecurity risks caused by human error and lack of awareness.

7. Incident Response Plan

An Incident Response Plan is a structured approach used to identify, manage, and recover from cybersecurity incidents. In a university environment, an effective incident response plan helps reduce the impact of cyberattacks, protect sensitive data, and restore normal operations quickly.

The incident response plan is designed to help the university detect, manage, and recover from

cybersecurity incidents effectively. CIA Triad alongwith Authentication are the pillars of our Incident Response Plan:

Confidentiality: *Protecting sensitive student, faculty, and university data from unauthorized access.*

Integrity:*Ensuring that university data and systems are accurate and not modified illegally.*

Availability: *Ensuring that university systems and online services remain accessible to authorized users.*

Authentication:Verifying the identity of users before granting access to university systems and resources.

7.1 Identification

The first step is to identify suspicious activities or cybersecurity incidents within the university system.

Activities

- Monitor failed login attempts and unusual user behavior.
- Detect malware alerts, phishing emails, or unauthorized access attempts.
- Analyze system logs and network activity for suspicious actions.

Objective

Early identification helps prevent cyber threats from spreading and causing major damage.

7.2 Containment

Once a threat is detected, immediate actions should be taken to limit its impact on university systems.

Activities

- Disconnect affected systems from the network if required.
- Block suspicious user accounts or IP addresses.
- Restrict access to compromised systems or files.

Objective

Containment helps stop the attack from spreading to other systems and reduces

further damage.

7.3 Eradication

This phase focuses on removing the cause of the cybersecurity incident.

Activities

- Remove malware or malicious files from affected systems.
- Fix security vulnerabilities and update affected software.
- Reset compromised passwords and strengthen security settings.

Objective

Eradication ensures that the threat is completely removed from the university environment.

7.4 Recovery

After the threat is removed, systems and services should be restored safely.

Activities

- Restore affected systems and data using backups.
- Verify that systems are functioning properly and securely.
- Monitor systems continuously after recovery.

Objective

Recovery helps resume normal university operations while ensuring systems remain secure.

7.5 Review and Improvement

After resolving the incident, the university should analyze the event and improve security measures.

Activities

- Document details of the incident and actions taken.
- Identify weaknesses that allowed the attack to occur.
- Update security policies and awareness programs if necessary.

Objective

This step helps improve future incident response and strengthens overall cybersecurity preparedness.

8. User Awareness Measures

User awareness plays an important role in improving cybersecurity within a university environment. Many cyberattacks occur because users are unaware of online threats such as phishing emails, weak passwords, and unsafe browsing practices. Proper awareness programs help students, faculty, and staff use university systems safely and responsibly.

1. Cybersecurity Awareness Programs

Regular workshops, seminars, and training sessions should be conducted to educate students and staff about common cyber threats and safe cybersecurity practices.

2. Phishing Awareness

Users should be trained to identify suspicious emails, fake websites, and fraudulent links that may attempt to steal login credentials or personal information.

3. Password Security Awareness

Students and staff should be encouraged to use strong and unique passwords and avoid sharing passwords with others. The use of Multi-Factor Authentication (MFA) should also be promoted.

4. Safe Internet and Wi-Fi Usage

Users should be educated about safe browsing practices and the risks of using unsecured public or university Wi-Fi networks without proper protection.

5. Incident Reporting Awareness

Students and staff should be encouraged to report suspicious activities, unknown emails, or cybersecurity incidents immediately to the university IT department for quick response and investigation.

9. Practical Implementation

Although this project is mainly theory-based, a sample cybersecurity awareness website was designed to demonstrate possible cyber threats and security risks faced by university systems. The website was created to provide information about common attacks on universities and the importance of cybersecurity policies, incident response, and user awareness.

The purpose of the sample website was to help users understand:

- Common cyber threats in universities
- Risks related to weak passwords and unauthorized access
- The importance of cybersecurity awareness
- Basic security practices to protect university systems

Features of the Sample Website

- Information about common cyberattacks such as phishing, malware, ransomware, and brute-force attacks
- Cybersecurity awareness tips for students and staff
- Security policy recommendations
- Risk assessment and attack-related information
- Basic overview of incident response measures

Technologies Used

- Linux Operating System
- HTML and CSS
- Python and Flask (for basic web functionality)

This implementation helped in presenting the theoretical concepts of cybersecurity risk assessment in a simple and interactive manner while improving understanding of cybersecurity threats in university environments.

10.Screenshots of our system



11. Advantages

● Helps identify major cybersecurity risks in university systems. ● Improves understanding of cybersecurity policies and risk management. ● Promotes safe cybersecurity practices among students and staff. ● Highlights the importance of incident response and continuous monitoring. ● Helps protect sensitive university data from unauthorized access and cyber threats.

- Encourages awareness about phishing attacks, weak passwords, and malware risks.
- Supports better security planning for university networks and online systems.
- Provides basic understanding of cybersecurity monitoring and threat detection concepts.

12. Limitations

- The project is mainly theoretical and does not provide complete real-time cybersecurity protection.
- Only basic cybersecurity risks and security measures are discussed.
- The practical is limited to a simple attack simulation concept.
- Advanced cybersecurity technologies such as AI-based threat detection and full intrusion detection systems are not included.
- The project does not cover large-scale enterprise-level security management.
- Continuous monitoring and maintenance would be required in a real university environment.
- Human errors and lack of user awareness can still create security risks despite security policies.

13. Future Scope

- Advanced cybersecurity technologies such as Intrusion Detection Systems (IDS) and AI-based threat detection can be implemented in the future.

- Real-time monitoring and automated alert systems can be added for faster threat detection and response.
- Multi-Factor Authentication (MFA) and advanced access control mechanisms can be integrated for improved security.
- Cloud-based security solutions and secure data backup systems can be implemented to improve data protection.
- Regular cybersecurity training and awareness programs can be expanded for students, faculty, and staff.
- Advanced risk assessment tools and network monitoring systems can be developed for better cybersecurity management.
- The system can be enhanced to handle large-scale university networks and more complex cyber threats.

14. Conclusion

This project focused on cybersecurity risk assessment in a university environment by identifying major cyber threats, studying their impact, and proposing suitable security policies and incident response measures. Universities are highly vulnerable to cyberattacks because they manage large amounts of sensitive student, faculty, and research data through interconnected digital systems.

The project also highlighted the importance of cybersecurity awareness, strong security policies, and continuous monitoring in protecting university networks and information systems. Proper implementation of cybersecurity measures can help reduce risks such as unauthorized access, phishing attacks, malware infections, and data breaches.

Overall, this project helped in understanding the importance of cybersecurity management and risk assessment in educational institutions and emphasized the need for secure and responsible use of digital technologies within university environments.

15. References

1. OWASP Foundation
2. National Institute of Standards and Technology (NIST) Cybersecurity Framework
3. A Systematic Review of Cybersecurity Risks in Higher Education 4. Understanding Cyber Threats Against Universities, Colleges, and Schools 5. Cyber Security and Universities – Managing the Risk
6. https://www.mdpi.com/1999-5903/13/2/39?